NIRAJ KUMAR GUPTA

Engineer & Lecturer — Networking, Routing & Switching | Cloud Computing | Computer Systems

Kathmandu, Nepal | guptaniraj2051@gmail.com

PROFESSIONAL SUMMARY

Senior Engineer at Nepal Doorsanchar Company Limited (NDCL) with 8+ years of hands-on experience across national telecommunications and networking infrastructure — routing, switching, IP addressing, network operations, and Cisco-based systems — combined with a Master's degree (First Division with Distinction) in Communication and Knowledge Engineering from the Institute of Engineering, Tribhuvan University. Complements deep industry networking practice with several years of visiting lecturer experience across colleges affiliated to the University of Sunderland (UK), Asia Pacific University (Malaysia), Tribhuvan University and Pokhara University (Nepal), and Purvanchal University, teaching Cloud Computing, Switching and Routing, Digital Logic, Data Communication, C Programming, Python Programming (CET140), and AI/ML. Brings live network-engineering context, real-world troubleshooting scenarios, and production infrastructure experience directly into Routing and Switching instruction.

RELEVANT NETWORKING & TELECOM EXPERIENCE

Senior Engineer **2018 – Present**

Nepal Doorsanchar Company Limited (NDCL), Nepal

- Provide technical expertise and operational support for critical national telecommunications infrastructure, including routing, switching, and network operations across production systems.
- Apply cloud computing (Microsoft Azure), networking, and embedded-systems expertise to live infrastructure projects, working directly with routing protocols, IP addressing, and network device configuration.
- Mentor graduate-level students during field visits, translating complex network and telecom systems — including routing and switching concepts — into accessible technical explanations for learners.

TEACHING EXPERIENCE

Gateway Professional College **2026 – Present**

Affiliated to Purvanchal University

- Digital Logic (BT102CO).

UniGlobe College **2026 – Present**

Affiliated to Pokhara University

- Switching and Routing (NCS 485).

Visiting Lecturer **2026 – Present**

ISMT College, Kathmandu — Affiliated to University of Sunderland, UK

- Delivered instruction and supervised student output for Specialist Projects (CET140).

Visiting Lecturer **2022 – 2025**

ISMT College, Pokhara — Affiliated to University of Sunderland, UK

- Delivered lectures and laboratory sessions for Cloud Computing (Unit 28), covering compute, storage, networking, and deployment on Microsoft Azure.
- Supervised and evaluated undergraduate research projects under module CET351 (Research).
- Delivered instruction and supervised student output for Specialist Projects (CET140).

Visiting Lecturer **2022 – 2025**

Infomax College, Pokhara — Affiliated to Asia Pacific University (APU), Malaysia

- Taught Software Systems & Computing Concepts (CT123-3-1) to undergraduate students.
- Demonstrated microcontroller and computer-architecture fundamentals using the Little Man Computer (LMC) simulator.

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Visiting Lecturer **2022 – 2025**

SOCH College of IT, Pokhara — Affiliated to Tribhuvan University

- Instructed students in CSC115 – C Programming, building programming logic and problem-solving skills from first principles.

EDUCATION

M.Sc. — Communication & Knowledge Engineering **2023**

Institute of Engineering (IOE), Western Region Campus, Tribhuvan University, Nepal | Result: First Division with Distinction

Coursework: Machine Learning, Pattern Recognition, Advanced Data Mining, Social Network Mining, Telecommunication & Computer Networks.

B.E. — Electronics & Communication Engineering **2016**

Institute of Engineering (IOE), Western Region Campus, Tribhuvan University, Nepal | Result: First Division

TECHNICAL SKILLS

- Networking & Telecom: Routing & Switching, Cisco networking, IP addressing, telecommunications infrastructure, network operations & troubleshooting
- Cloud: Microsoft Azure (compute, storage, deployment)
- AI / ML / Deep Learning: Generative Adversarial Networks (GANs), pattern recognition, large-scale data mining
- Hardware & Embedded Systems: Arduino prototyping, RFID, biometric and multi-factor authentication systems
- Programming & Tools: Python (data science & ML pipelines), C (systems programming), Microsoft Office

RESEARCH & PUBLICATIONS

Progressive Face Ageing via Generative Adversarial Networks (GAN)

- Master's thesis. Designed and implemented a GAN-based model to synthesize realistic, age-progressed facial images. Published on ResearchGate (researchgate.net/publication/370580661).

Triple-Layer Security System (Major Project)

- Multi-factor authentication vault built on Arduino, integrating RFID, PIN entry, and biometric fingerprint verification across three independent security layers.

RFID Access Control System (Minor Project)

- Contactless access-control system built using Arduino Uno and Mifare RC522 RFID modules.

LANGUAGES

English (Professional) · Nepali (Native) · Hindi (Fluent) · Bhojpuri (Fluent)